

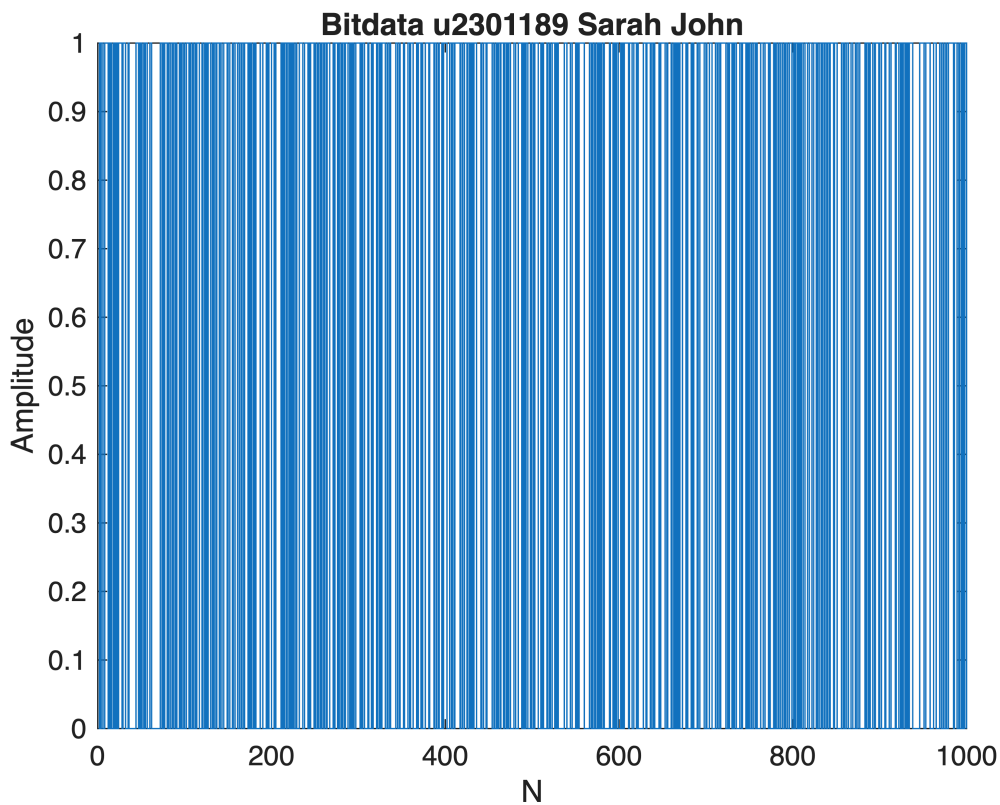
Error performance of BPSK

Generation of BPSK

```
clc
clear all
close all
N=input("Enter the number of Bits:");
bit=randi([0,1],1,N)
```

```
bit = 1×1000
      0      1      0      1      1      0      0      1      1      1      1      0      1...
```

```
stairs(bit)
xlabel("N");
ylabel("Amplitude");
title("Bitdata u2301189 Sarah John");
```

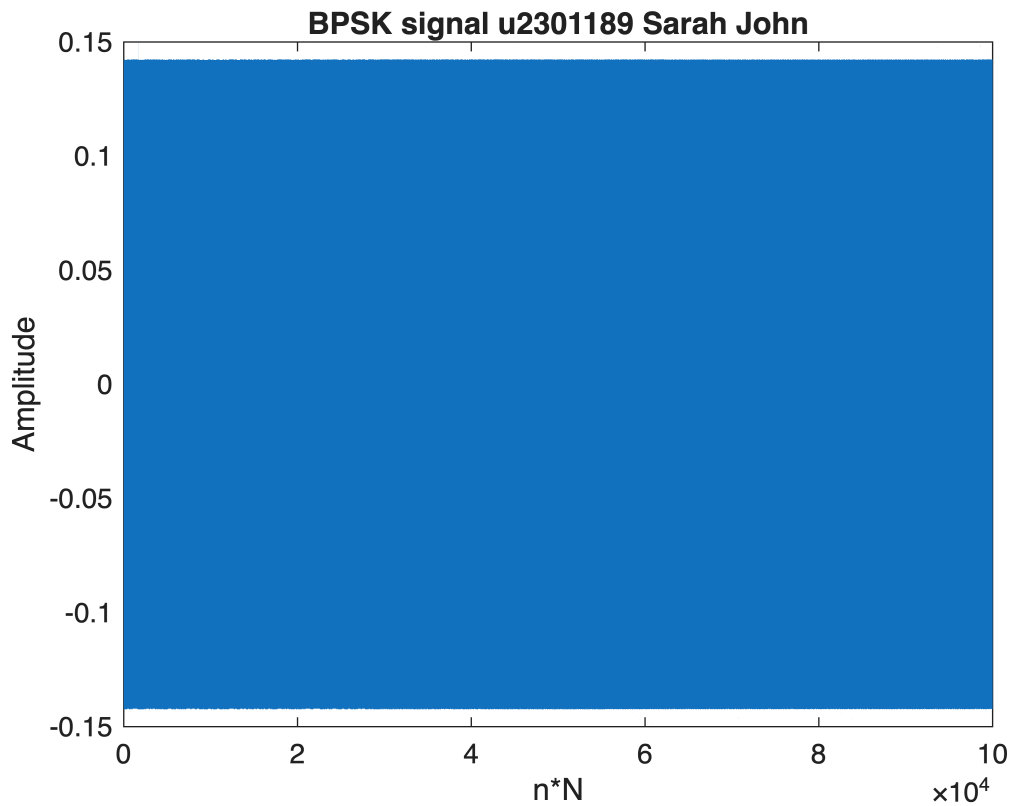


```
A=input("Enter the amplitude of sine:");
n=input("Enter the number of samples:");
t=linspace(0,4*pi,n);
c=A*sin(t);
c_t=transpose(c);
E=c*c_t;
cnorm=c/sqrt(E);
bpsk=[];
for i=1:N
if bit(i)==1
```

```

bpsk=[bpsk cnorm];
elseif bit(i)==0
bpsk=[bpsk cnorm*-1];
end
end
plot(bpsk)
xlabel("n*N");
ylabel("Amplitude");
title("BPSK signal u2301189 Sarah John");

```

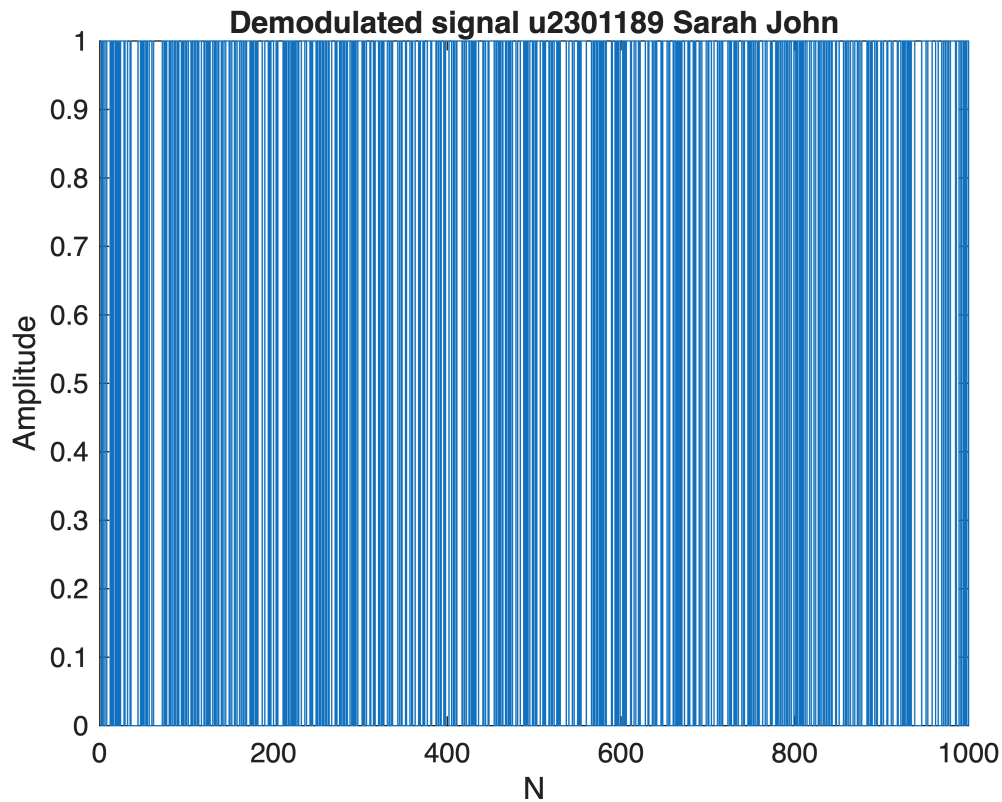


Demodulation of BPSK

```

rb=reshape(bpsk,[n,N]);
product=cnorm*rb;
deconst=[];
for j=1:N
if product(j)>0
deconst=[deconst 1];
elseif product(j)<0
deconst=[deconst 0];
end
end
stairs(deconst)
xlabel("N")
ylabel('Amplitude')
title("Demodulated signal u2301189 Sarah John")

```



Noise addition and Demodulation

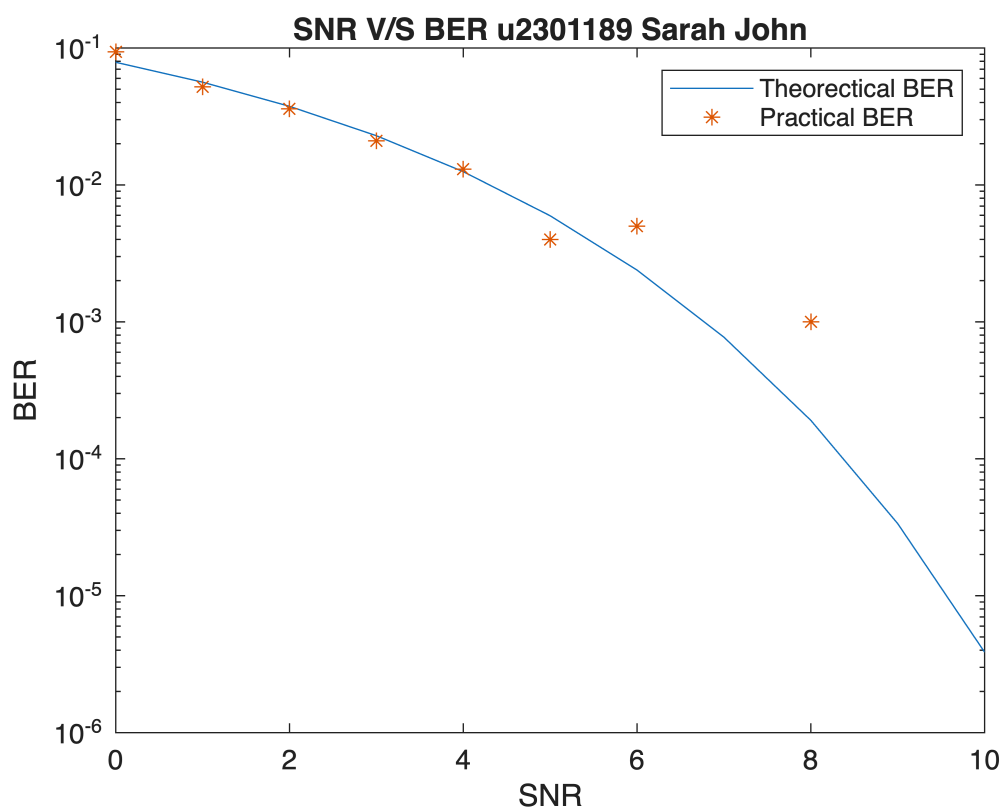
```
th_BER=[];
pr_BER=[];
SNR_array=[];
for SNR=0:1:10
    SNR_array=[SNR_array SNR];
    No=1/10^(0.1*SNR);
    t_BER=qfunc(sqrt(2/No));
    th_BER=[th_BER t_BER];
    var=No/2;
    std_dev=sqrt(var);
    Noise=std_dev*complex(randn(1,length(bpsk)),randn(1,length(bpsk)));
    bpsk_channel=bpsk+Noise;
    bprb=reshape(bpsk_channel,[n,N]);
    product1=cnorm*bprb;
    deconst1=[];
    for k=1:N
        if product1(k)>0
            deconst1=[deconst1 1];
        elseif product1(k)<0
            deconst1=[deconst1 0];
        end
    end
```

Error Calculation

```

Error=0;
for l=1:N
if bit(l)~=deconst1(l)
Error=Error+1;
end
end
BER=Error/N;
pr_BER=[pr_BER BER];
end
semilogy(SNR_array,th_BER,SNR_array,pr_BER,"*");
xlabel("SNR")
ylabel("BER")
legend("Theoretical BER","Practical BER")
title("SNR V/S BER u2301189 Sarah John")

```



Scatter Plot

```

I_noisy = real(cnorm * bprb);
Q_noisy = imag(cnorm * bprb);
scatter(I_noisy, Q_noisy, 'blue', 'filled')
xlabel('In-phase'); ylabel('Quadrature');
title('BPSK Constellation (Scatter Plot) u2301189 Sarah John')

```

